

SAFFORD, AZ, USA

WMO: 722747

Lat: 32.855N

Lon: 109.635W

Elev: 968

StdP: 90.22

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 93084

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-3.1	-1.5	-19.7	0.7	14.1	-17.5	0.9	10.7	11.8	10.1	10.6	10.9	1.6	110	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	40.8	18.2	39.2	18.0	37.9	18.0	21.6	31.0	21.1	30.6	20.7	30.3	5.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	16.2	23.1	18.9	15.5	23.3	18.0	14.6	24.2	67.3	30.5	65.6	30.3	63.8	30.0	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.3	8.3	DB	-6.8	42.8	2.4	0.8	-8.6	43.4	-10.0	43.8	-11.3	44.3	-13.0	44.9	(4)
(5)			WB	-8.7	22.7	2.5	1.0	-10.5	23.4	-11.9	23.9	-13.3	24.5	-15.1	25.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.3	8.6	10.8	14.4	18.3	23.2	29.0	30.1	29.0	26.2	19.9	13.4	8.4		
	DBStd	8.46	3.23	3.81	3.43	3.66	3.49	2.84	2.43	2.29	2.51	3.52	3.92	3.21		
	HDD10.0	187	64	31	6	1	0	0	0	0	0	1	14	70		
	HDD18.3	1174	301	212	126	45	5	0	0	0	0	24	153	309		
	CDD10.0	3582	21	53	143	249	408	569	622	589	485	308	116	19		
	CDD18.3	1528	0	0	4	43	154	319	364	331	235	73	4	0		
	CDH23.3	21752	0	24	236	867	2456	4988	5018	4291	2819	956	97	0		
	CDH26.7	12332	0	2	42	321	1320	3205	3078	2502	1487	363	12	0		
(14) Wind	WSAvg	3.5	3.0	3.2	3.6	4.0	3.9	4.0	4.0	3.5	3.1	3.1	3.0	3.0		
(15) Precipitation	PrecAvg	240	18	18	14	6	5	7	38	43	29	24	14	23		
	PrecMax	441	79	67	43	36	58	36	92	112	107	112	61	120		
	PrecMin	120	0	0	0	0	0	0	6	2	0	0	0	0		
	PrecStd	72	17	17	13	8	10	9	20	28	27	26	14	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.2	26.2	30.0	34.0	38.9	42.6	42.3	40.7	37.9	34.0	28.1	22.0		
		MCWB	9.3	10.8	12.4	13.9	16.6	18.3	19.0	19.2	18.1	16.2	12.5	9.8		
	2%	DB	19.9	23.6	27.7	32.0	36.8	41.0	40.8	39.0	36.8	32.3	26.1	19.8		
		MCWB	8.4	9.8	11.6	13.0	15.6	17.5	18.8	19.2	17.7	15.4	12.1	9.1		
	5%	DB	18.0	21.7	26.2	30.3	35.0	39.5	39.0	37.6	35.2	30.6	24.0	17.6		
		MCWB	7.8	9.1	10.9	12.5	14.7	16.9	18.8	19.2	17.5	14.7	11.2	8.1		
	10%	DB	16.3	19.3	24.1	28.4	33.3	38.0	37.5	36.2	33.8	28.7	22.1	15.8		
		MCWB	7.0	8.3	10.1	12.0	14.2	16.8	18.8	19.2	17.4	14.2	10.5	7.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	12.5	13.9	15.8	18.4	20.9	22.3	22.4	21.3	19.1	14.9	12.1		
		MCDB	16.2	20.7	25.2	30.5	35.7	35.2	31.2	31.4	29.3	26.6	22.2	17.2		
	2%	WB	10.1	11.3	12.6	14.4	16.8	19.9	21.6	21.6	20.5	17.5	13.5	10.5		
		MCDB	16.7	19.4	24.6	29.0	33.7	33.9	30.7	31.1	28.8	26.6	22.4	16.7		
	5%	WB	9.0	10.4	11.6	13.2	15.8	19.0	21.1	21.1	19.9	16.5	12.3	9.3		
		MCDB	15.9	18.7	23.6	27.8	32.3	33.5	30.3	30.9	28.4	26.3	21.6	15.6		
	10%	WB	7.9	9.2	10.7	12.2	14.9	18.2	20.7	20.7	19.2	15.5	11.2	8.1		
		MCDB	14.7	17.7	22.4	26.8	31.1	33.6	30.0	30.5	28.3	25.1	20.4	14.5		
(35) Mean Daily Temperature Range	MDBR		14.4	15.4	17.0	17.9	18.2	17.7	14.2	13.9	14.6	16.0	15.3	13.9		
	5% DB	MCDBR	17.1	19.2	20.0	20.4	20.0	19.0	16.4	15.5	16.4	18.1	17.9	16.5		
		MCWBR	9.2	9.7	9.3	9.6	8.9	7.4	5.1	4.2	5.5	7.6	8.8	8.9		
	5% WB	MCDBR	14.0	16.0	17.9	18.7	18.8	16.2	13.3	13.3	12.4	13.9	15.1	14.1		
		MCWBR	7.9	8.2	8.5	9.0	9.0	6.4	3.3	3.4	4.0	5.7	7.6	7.8		
(40) Clear-Sky Solar Irradiance	taub		0.256	0.260	0.282	0.299	0.318	0.336	0.389	0.372	0.333	0.290	0.265	0.255		
	taud		2.668	2.667	2.571	2.500	2.484	2.440	2.417	2.432	2.533	2.618	2.646	2.665		
	Ebn at Noon		988	1012	1003	990	966	943	891	905	934	962	967	971		
	Edh at Noon		72	80	95	107	110	114	117	113	98	83	73	68		
(44) All-Sky Solar Radiation	RadAvg		3.47	4.40	5.90	7.26	8.07	8.10	6.80	6.38	5.82	4.93	3.87	3.18		
	RadStd		0.29	0.40	0.33	0.30	0.38	0.46	0.36	0.21	0.29	0.37	0.21	0.17		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page